

ROGERS, AR, USA

WMO: 723449

Lat: 36.372N

Lon: 94.107W

Elev: 412

StdP: 96.47

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 53954

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.2	-9.1	-17.7	0.8	-10.2	-15.0	1.1	-6.7	11.7	13.7	10.7	11.3	3.1	10	0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	34.9	23.3	33.2	23.6	32.2	23.3	25.4	31.4	24.8	30.7	24.1	29.8	4.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.6	19.4	28.9	22.9	18.6	28.1	22.4	18.0	27.5	80.0	31.4	77.2	30.7	74.7	29.9	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.1	8.7	7.8	-15.7	36.4	2.1	2.1	-17.2	37.9	-18.4	39.1	-19.6	40.3	-21.2	41.8
			-16.5	26.0	2.3	0.8	-18.2	26.6	-19.5	27.1	-20.7	27.6	-22.4	28.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	2.6	4.9	9.2	14.2	18.8	23.7	25.9	25.4	21.6	15.4	9.4	4.0
	DBStd	9.49	6.46	6.13	5.84	4.76	4.24	3.21	2.91	2.83	4.17	4.82	5.30	5.72
	HDD10.0	787	240	162	84	17	2	0	0	0	1	11	73	198
	HDD18.3	2183	487	376	287	137	47	3	0	1	18	115	270	443
	CDD10.0	2482	11	20	58	143	274	411	493	477	347	178	55	14
	CDD18.3	837	0	1	3	13	62	164	236	218	114	24	2	0
	CDH23.3	7706	0	3	19	84	454	1491	2429	2084	963	173	7	0
	CDH26.7	3058	0	1	1	6	99	559	1090	918	357	28	0	0
(14) Wind	WSAvg	3.6	3.9	4.0	4.2	4.4	3.6	3.3	2.9	2.8	2.9	3.3	3.8	3.7
(15) Precipitation	PrecAvg	1154	64	57	92	113	150	119	97	70	108	84	108	79
	PrecMax	1664	182	151	197	317	300	312	268	154	263	177	247	325
	PrecMin	824	4	8	20	39	37	27	24	0	13	26	14	8
	PrecStd	205	45	39	42	64	57	72	55	37	58	37	69	62
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.9	22.3	25.7	27.4	31.0	34.7	37.3	37.3	35.0	29.1	24.0	20.3
		MCWB	13.1	15.3	17.0	18.1	21.5	23.6	22.8	22.7	22.2	21.1	17.1	15.3
	2%	DB	17.6	19.1	22.9	25.9	29.0	32.9	35.0	34.6	32.3	27.3	21.4	17.7
		MCWB	12.7	13.0	15.0	17.4	21.1	23.6	23.6	23.0	22.3	19.3	15.5	13.3
	5%	DB	15.1	17.2	20.9	24.0	27.6	31.4	33.4	32.9	30.7	25.3	19.5	15.7
		MCWB	10.5	12.1	14.0	16.6	20.5	23.1	24.1	23.2	21.9	17.9	14.7	11.7
	10%	DB	12.6	14.9	18.7	22.4	26.1	29.9	32.1	31.6	28.7	23.0	17.7	12.8
		MCWB	8.6	10.1	13.0	15.8	19.9	22.8	23.9	22.9	21.0	17.3	12.9	9.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.8	16.8	18.3	20.2	23.5	25.6	26.3	26.0	24.5	22.5	19.1	16.4
		MCDB	18.4	20.5	24.1	25.1	28.5	31.9	32.4	32.0	31.0	27.5	22.4	19.3
	2%	WB	13.9	14.9	16.7	19.0	22.4	24.8	25.6	25.1	23.5	21.0	17.1	14.2
		MCDB	16.4	17.9	21.2	23.7	27.4	30.6	31.7	31.0	29.8	25.1	20.1	16.9
	5%	WB	11.5	12.9	15.2	18.0	21.4	24.0	25.0	24.3	22.8	19.7	15.4	12.3
		MCDB	14.0	16.1	19.5	22.2	26.2	29.5	31.0	30.3	28.5	23.2	18.6	14.7
	10%	WB	8.8	10.5	13.5	16.9	20.5	23.2	24.4	23.6	21.9	18.2	13.8	10.0
		MCDB	12.1	14.5	17.6	21.0	25.0	28.5	30.0	29.4	26.6	22.3	16.9	12.4
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	10.3	10.9	11.2	10.4	10.0	10.1	10.5	10.7	11.1	10.4	9.3
		MCDBR	12.8	13.2	13.6	12.5	11.3	11.2	11.5	12.2	12.0	12.3	11.9	12.1
	5% WB	MCWBR	8.7	8.9	8.1	6.9	5.5	4.4	3.8	4.1	4.4	6.1	7.8	8.3
		MCWBR	10.9	11.4	11.3	10.4	10.2	10.3	10.3	10.3	10.7	10.2	10.7	10.5
(40) Clear-Sky Solar Irradiance	taub		0.301	0.304	0.338	0.376	0.417	0.420	0.442	0.442	0.384	0.342	0.311	0.290
		taud	2.487	2.486	2.425	2.329	2.264	2.281	2.273	2.248	2.381	2.484	2.523	2.536
	Ebn at Noon		898	939	930	906	869	863	841	833	869	883	881	891
		Edh at Noon	82	92	108	125	136	133	134	134	112	92	79	73
(44) All-Sky Solar Radiation	RadAvg		2.45	2.99	4.02	5.08	5.63	6.44	6.35	5.77	4.93	3.67	2.63	2.09
	RadStd		0.21	0.42	0.24	0.29	0.47	0.55	0.39	0.40	0.38	0.47	0.29	0.26

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page